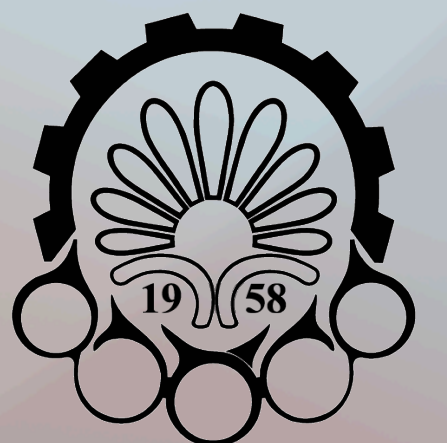


Deep Learning Meets Vision: A Deep Dive into CNNs!

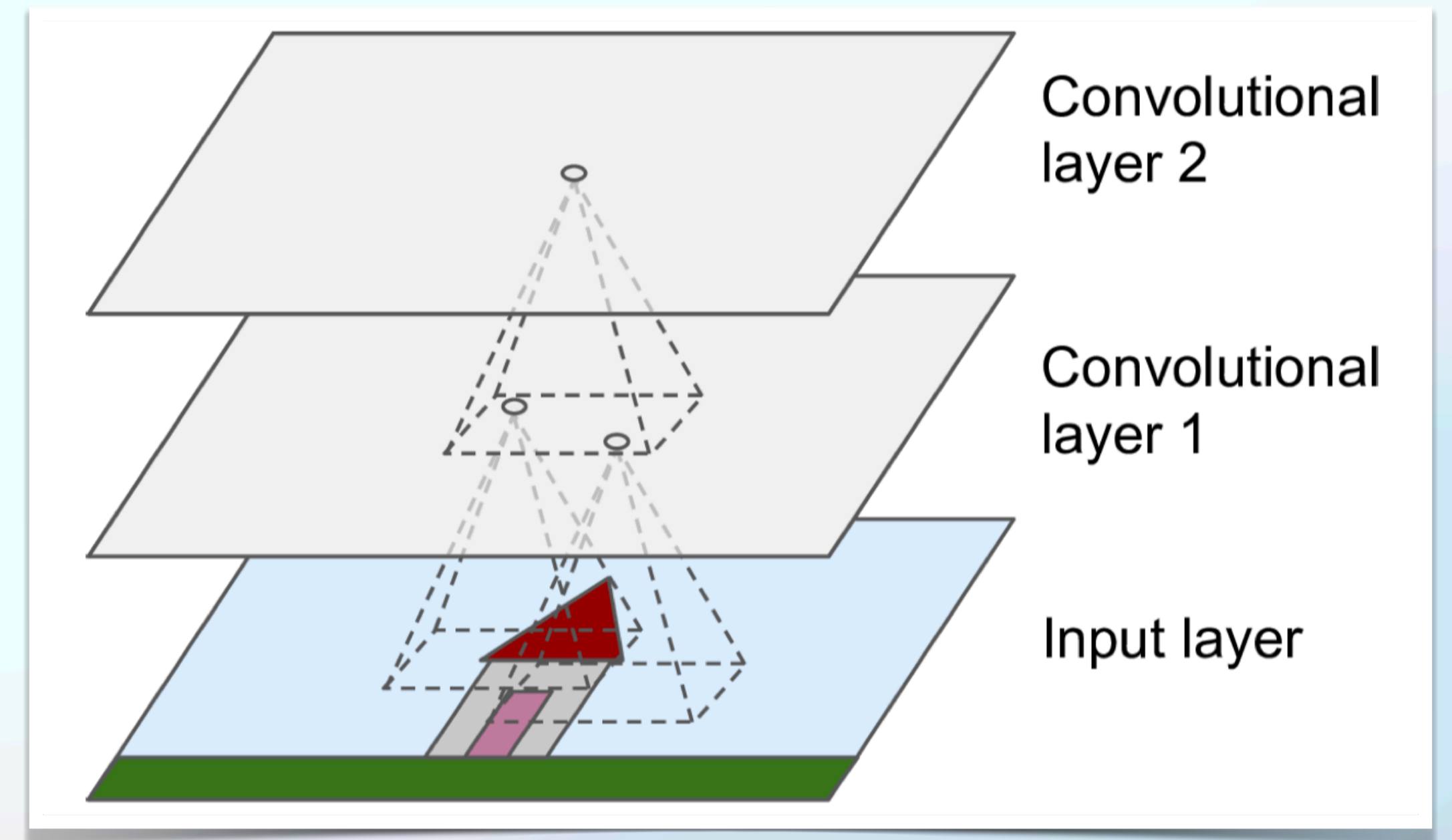
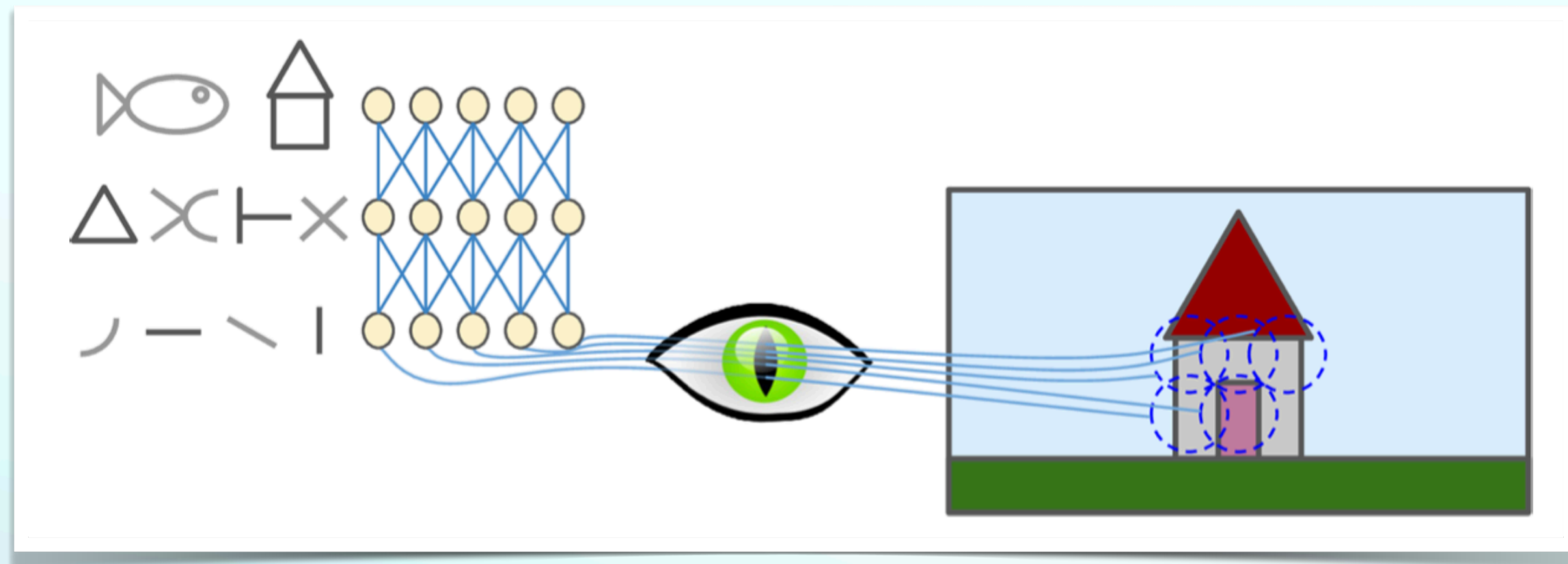
Computation Intelligence and its Application in Mechatronics

Winter 2025



Amirkabir University of Technology
(Tehran Polytechnic)

CNN Fundamentals



CNN Architecture Overview

Key Components of CNNs

- Convolutional Layers
- Activation Functions
- Pooling Layers
- Fully Connected Layers

Convolutional Layer

convolutional filter

0	1	0
1	0	1
0	1	0

128	97	53	201	198
35	22	25	200	195
37	24	28	197	182
33	28	92	195	179
31	40	100	192	177

181		

128	97	53	201	198
35	22	25	200	195
37	24	28	197	182
33	28	92	195	179
31	40	100	192	177

181	303	

128	97	53	201	198
35	22	25	200	195
37	24	28	197	182
33	28	92	195	179
31	40	100	192	177

181	303	618

128	97	53	201	198
35	22	25	200	195
37	24	28	197	182
33	28	92	195	179
31	40	100	192	177

181	303	618
115		

128	97	53	201	198
35	22	25	200	195
37	24	28	197	182
33	28	92	195	179
31	40	100	192	177

181	303	618
115	338	

128	97	53	201	198
35	22	25	200	195
37	24	28	197	182
33	28	92	195	179
31	40	100	192	177

181	303	618
115	338	605

128	97	53	201	198
35	22	25	200	195
37	24	28	197	182
33	28	92	195	179
31	40	100	192	177

181	303	618
115	338	605
169		

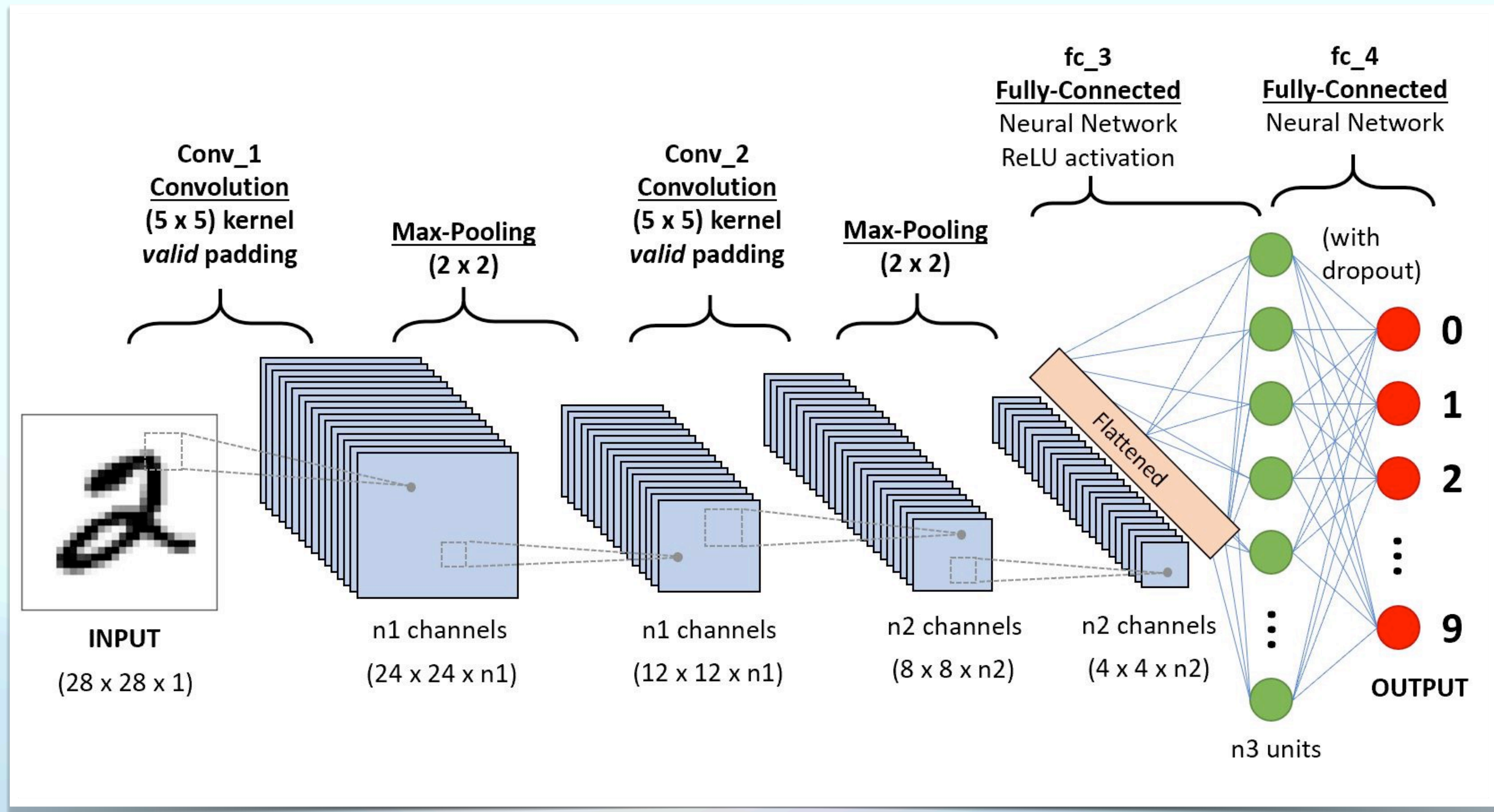
128	97	53	201	198
35	22	25	200	195
37	24	28	197	182
33	28	92	195	179
31	40	100	192	177

181	303	618
115	338	605
169	351	

128	97	53	201	198
35	22	25	200	195
37	24	28	197	182
33	28	92	195	179
31	40	100	192	177

181		

CNN Architecture Overview



CNN in Keras

convolutional Layers

1. 1D Convolution: `tf.keras.layers.Conv1D`:

- Used for processing sequential data such as time series, audio, and NLP tasks.
- Operates by sliding a 1D filter over the input sequence

2. 2D Convolution: `tf.keras.layers.Conv2D`

- Used for processing images (height, width, channels).
- Filters slide over 2D spatial data.

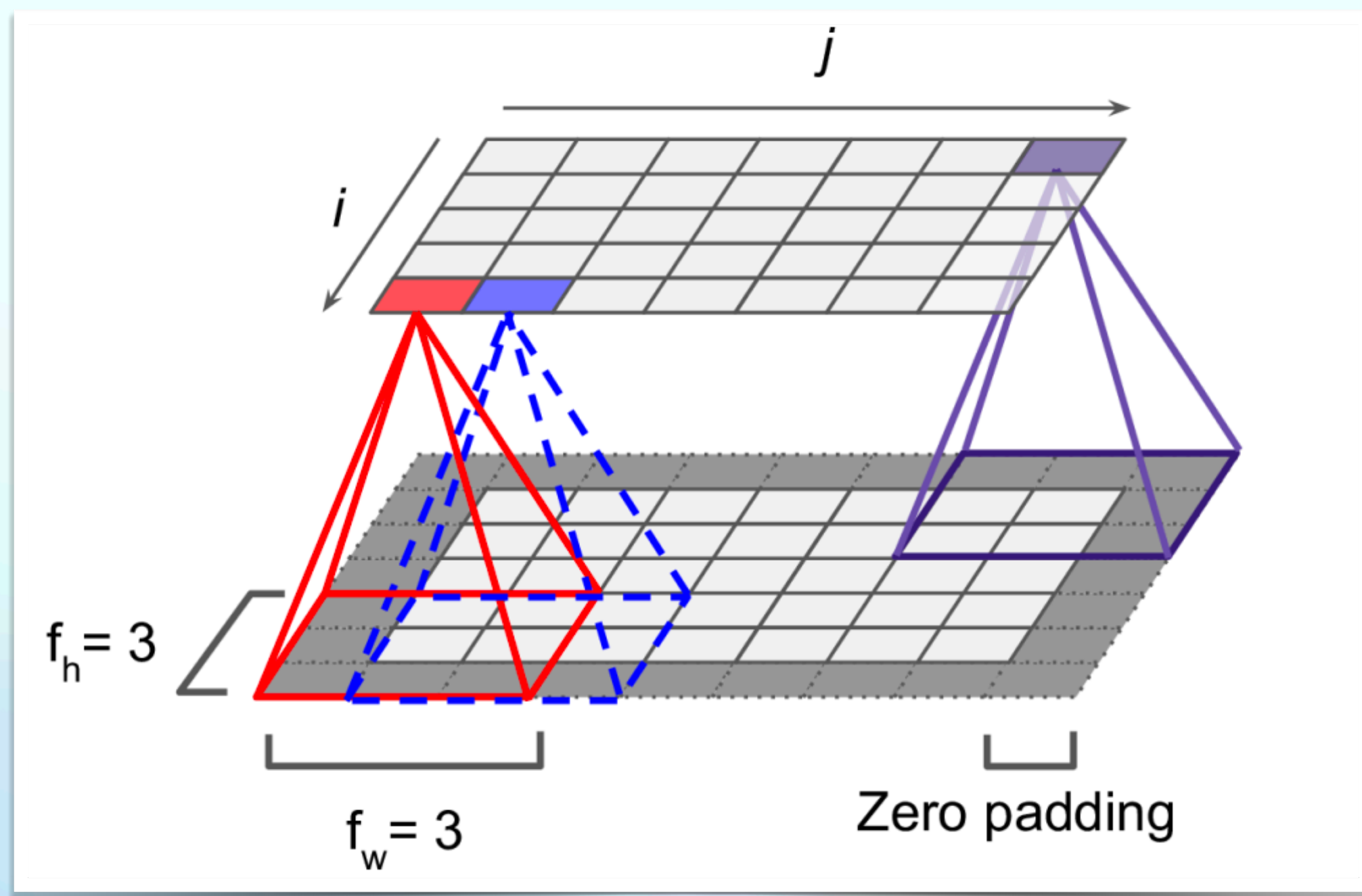
3. 3D Convolution: `tf.keras.layers.Conv3D`

- Used for volumetric data like MRI scans, video frames, and 3D object detection.
- Filters slide over three dimensions (depth, height, width).

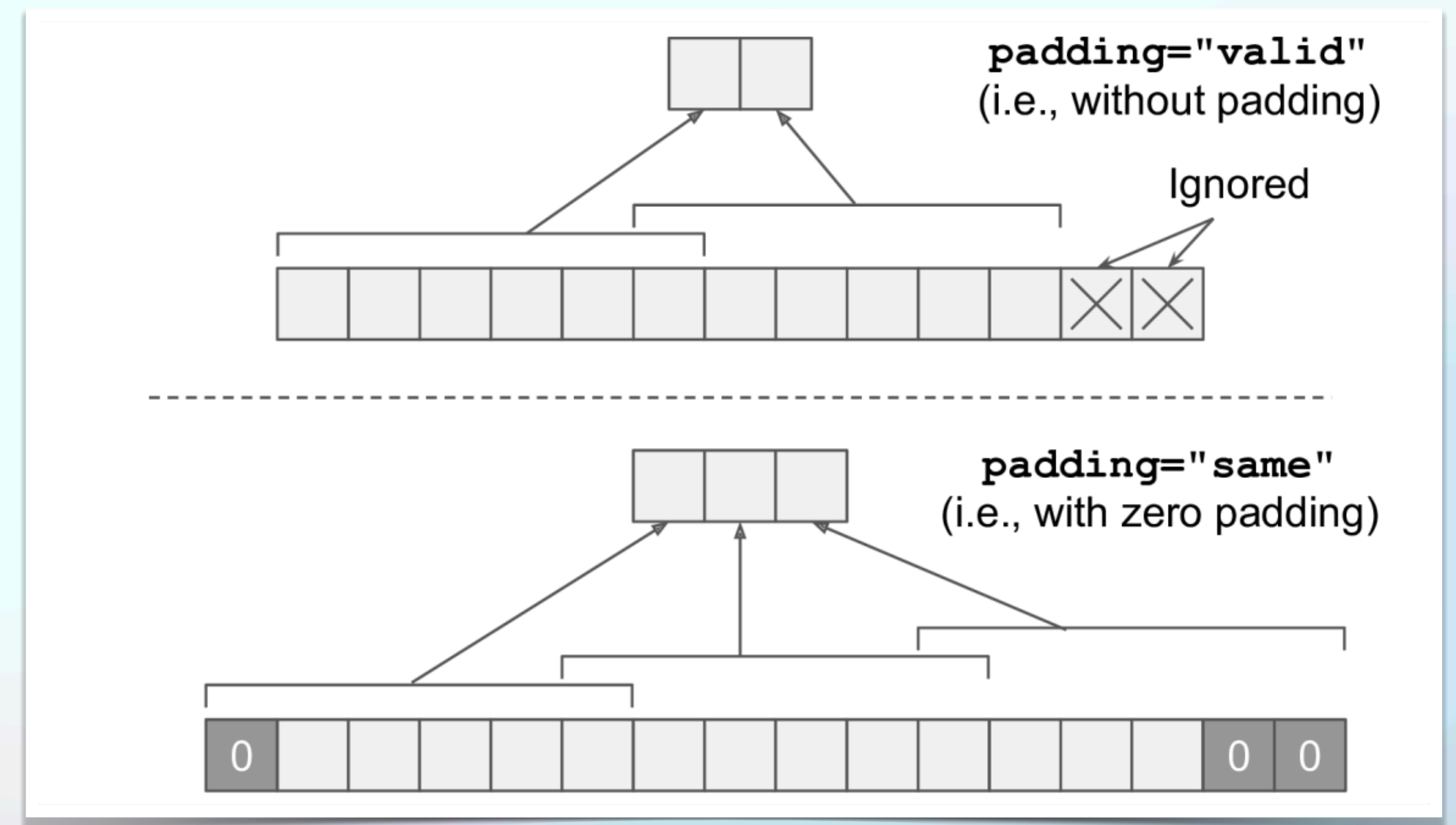
CNN in Keras

convolutional Layers

Kernel and stride size and

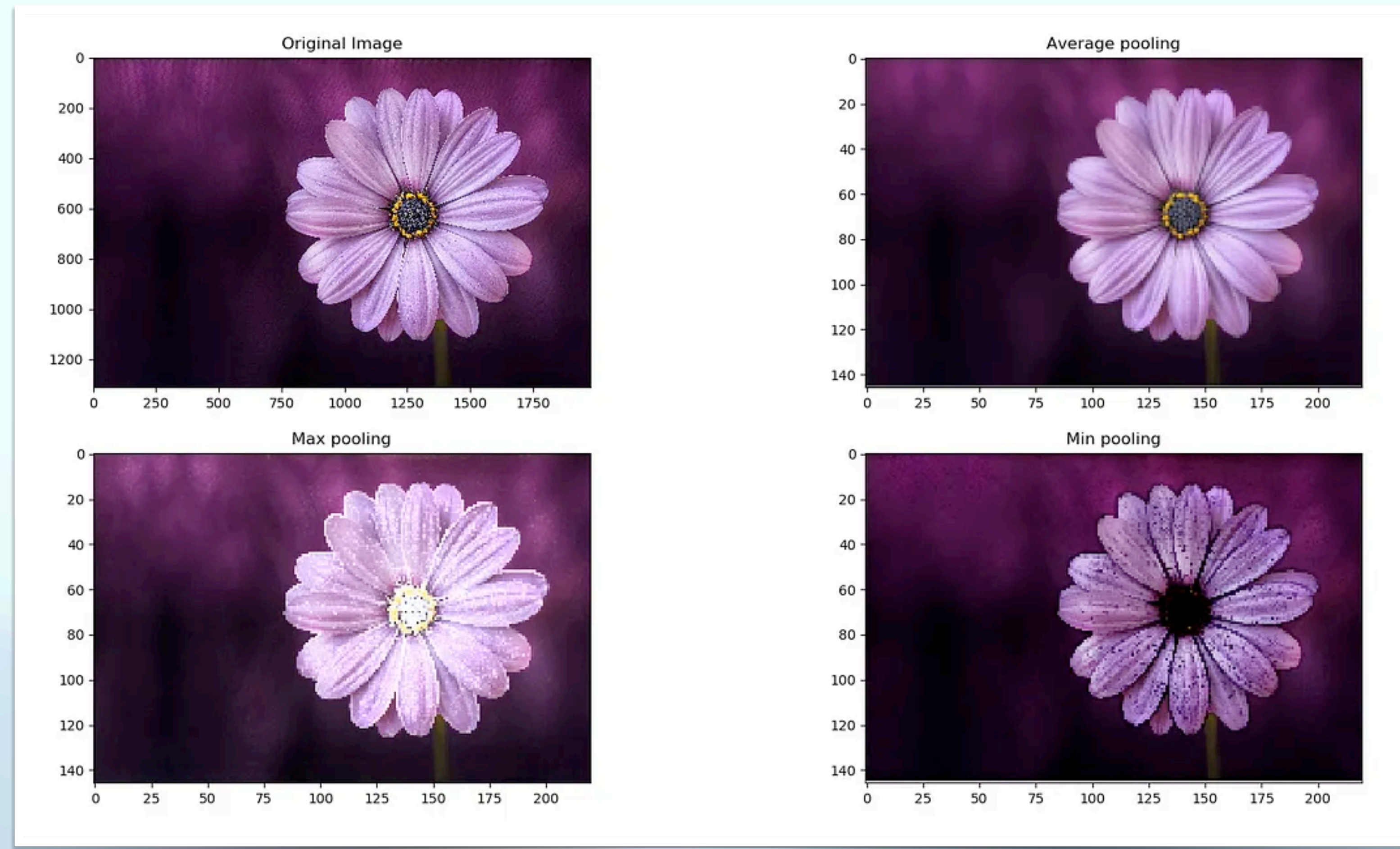


Padding



CNN in Keras

Pooling Operations



CNN in Keras

Pooling Operations

1. Average Pooling:

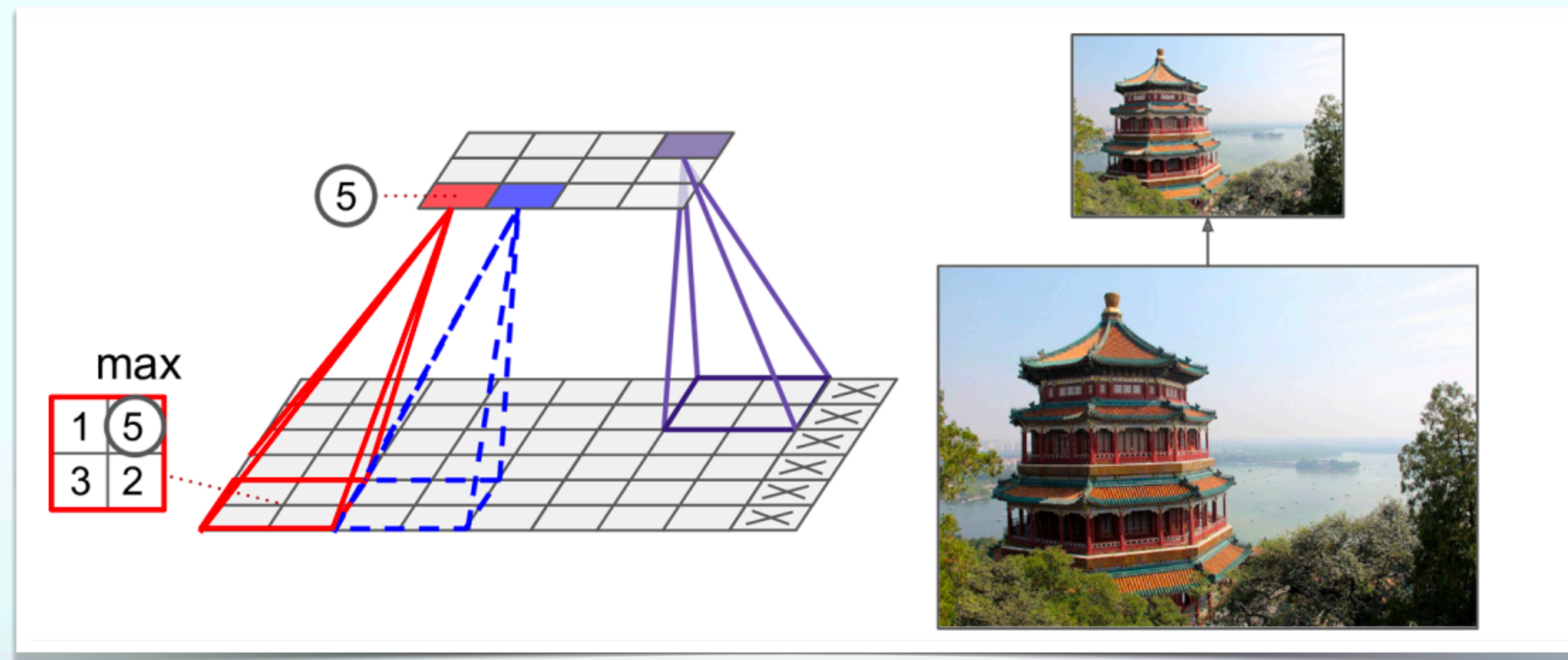
- `tf.keras.layers.AveragePooling1D`
- `tf.keras.layers.AveragePooling2D`
- `tf.keras.layers.AveragePooling3D`

2. Max Pooling

- `tf.keras.layers.MaxPool1D`
- `tf.keras.layers.MaxPool2D`
- `tf.keras.layers.MaxPool3D`

CNN in Keras

Pooling Operations



CNN in Keras

Pooling Operations

Average Pooling

2	3	1	4
5	6	7	8
9	10	11	12
13	14	15	16



4	5
11.5	13.5

Global Pooling

2	3	1	4
5	6	7	8
9	10	11	12
13	14	15	16



8.5

What's Comming Next...

- Introducing well-known architectures
- Diving deeper into CNN implementation